Format for B.Eng and M.Eng Research Projects and Dissertations in NAHPI

With additional notes from page 21

NB: This document complements, and does not in any way contradicts or supercedes or substitutes The University of Bamenda thesis and dissertation guide

1. Introduction to this document

The University of Bamenda through the publication of its Thesis and Dissertation guide (UBa, 2018) offers among other uniform standards regarding formats of Dissertations and Theses and allows flexibility to satisfy accepted practices in different domains in each Establishment. In accordance with the above and to ensure adherence to technical matters in the preparation and submission of Undergraduate Projects and Dissertations for degrees and certificates, this guide shall serve for the preparation of Undergraduate Projects and Dissertations for degree programs in the National Higher Polytechnic Institute (NAHPI) of The University of Bamenda.

2. Format of the End of Course B.Eng Project, MEng and MSc Dissertations

The entire main text (from Chapter 1 to Appendices) shall be **50 – 100 pages** for B.Eng project and **80 – 150 pages** for M.Eng and M.Sc dissertations depending on the nature of the topic and discipline. Otherwise, the School Scientific Committee (with external expert advice if necessary) will decide on the length of the Dissertation.

The composition of a Dissertation format shall be:

2.1 Preliminary pages

The elements of this section shall follow the order, each beginning on a new page:

- Cover page (Soft binding before defence and Hard or cardboard binding after defence)
- Title page
- Copyright page (Candidates hold copyright to their Dissertations and Theses regardless of whether they include a copyright page in them or register with the copyright office)
- Declaration (used before and after defence)
- Certification (used after defence)
- Acceptance page (required for postgraduate degrees) – this page confirms the Committee’s approval and acceptance of the Dissertation or Thesis
- Abstract (separate page and no longer than one page, Keywords (maximum 6) at the base)
- *Résume* (French translation of the Title on top followed by that of the Abstract)
- Dedication
- Acknowledgment
- Table of contents

- Lists of Tables
- Lists of Figures
- List of abbreviations

For the Cover and Title pages: A title of **at most 15 words** describing the Dissertation shall be presented. The title must be succinct, grammatically perfect; different from a textbook type title, and *should not end in a full-stop since it is not a sentence*.

Thus, instead of writing: “The Bamenda-Bambili Road in NW Cameroon”, which could be a textbook or a review article title, consider writing “The conception and design of a double-carriage road linking Bamenda and Bambili in NW Cameroon”

Thus, instead of writing: “Immuno-assay for breast cancer diagnosis”, consider writing “The development and characterisation of a novel sandwich immuno-assay for breast cancer diagnosis”

2.2 Specifications to be respected for Dissertations

2.2.1 **Language:** Dissertations in the National Higher Polytechnic Institute must be written in English. The use of **British English is compulsory (**For example; use **colour** instead of **color**).

2.2.2 **Paper size**: A4 Portrait. (**NB:** For the presentation of large Tables and Figures**,** Landscape may be used for pages containing such Tables and Figures**.)**

Top margin: 2.5 cm

Bottom margin: 2.5 cm

Left margin: 3.5 cm

Right margin: 2.5 cm.

2.2.3 Formatting text:

Font type: Times New Roman

Font size: 12

Line spacing: 1.5

Alignment: Justified

2.2.4 **Cover page colour**: The colour for binding of ALL Dissertations in the National Higher Polytechnic Institute shall be SKY BLUE.

2.2.5 **Pagination**: The page number shall be at the centre of the bottom of the page.

The preliminary part / aspect of the document shall be paginated in Roman numerals (i, ii, iii, iv, v,……). The rest of the document starting from the CHAPTER 1: INTRODUCTION page shall be paginated in Arabic numerals (1, 2, 3, 4, 5, ….) till the end of the book. Each Chapter shall commence on a separate page.

2.3 Tables, Figures, Formulae and Numberings:

Table Captions shall be placed **above Tables** at the left side. Figure captions shall be **placed below Figures** at the left side.

- Every Chapter, Table, Figure and equation shall be numbered in Arabic numerals and chronologically using chapter numbers, e.g. Table 3.1 for Table 1 under Chapter 3. Figure 2.4 standing for Figure 4 under Chapter 2. In the main text, the names of figures and tables must begin with capital letters, i.e. Figure… Table… e.g. "It can be been in Figure 3.2 that….." " Looking at Table 3.7….."
- Photographs and maps must be of high quality.

- If a Table or Figure continues onto a subsequent page(s), include the following caption: Table xxx (continued) or Figure xxxxx (continued).

2.4 Numbering of headings and subheadings

Heading: 1,

First subheading: 1.1

Second Subheading: 1.1.1

The numbering of the chapters and sub-headings shall be according to levels as follows:

E.g. Chapter 1: INTRODCUTION

1.1

    1.1.1

    1.1.2

1.2

    1.2.1

    1.2.2

    etc.

NB: The use of A, B or a, b, or 1a, 1b, to label headings or subheadings is specifically prohibited.

2.5 Headers or footers Students should not put any information as headers or footers

2.6 Binding: Use spiral binding, protected by a sheet of transparent paper before the defence and after the defence use wet binding.

3 Main Body of the Dissertation

Chapter 1: Introduction

Chapter 2: Literature Review

Chapter 3: Materials and Methods

Chapter 4: Results and Discussion

4.1. Results

4.2. Discussion

Chapter 5: Conclusions and Recommendations

5.1. Conclusions

5.2. Recommendations

References (does not carry a chapter number): *See detailed treatment of this below.*

Appendices (does not carry a chapter number)**:** This section shall appear at the end of the main body of the Dissertation (after the References). In the Appendix section, the appendices shall be announced in the same chronological order as they are made referenced to in the text or previous parts. It shall contain supplementary materials that are not essential parts of the text but which may be helpful in providing a more comprehensive understanding of the Dissertation. It may also be information that is cumbersome to be included in the main body of the Dissertation. Each Appendix shall start on a new page and shall be numbered.

4. DETAILS OF SOME ELEMENTS IN THE STRUCTURE OF DISSERTATION

4.1 Dedication

This section contains person or group of persons in whose honour the dissertation is written. It is optional.

4.2 Acknowledgments

Always starts with your supervisors

Then anyone who helped you:

1. Technically (including materials, supplies)

2. Intellectually (assistance, advice)

3. Financially (for example, Departmental support, travel grants)

The first person to be acknowledged must be your supervisor, then the co-supervisor, if any. Never forget to acknowledge the laboratory where you worked, as well as your HOD and lecturers.

4.3 Abstract

- Length should be one (1) paragraph of maximum 300 words for B.Eng projects, M.Eng and MSc Dissertations.

- The abstract must contain elements of: **the problem statement**; the aim or overall objective of the study or **purpose statement**; the **methodology**; the **key results obtained**; the **major conclusion** from the results.

- State three (3) to five (5) keywords.

Table of Contents

- list all headings and subheadings with page numbers

- indent subheadings

List of Figures

List page numbers of all figures. The list should include a **short title** for each figure but not the description of whole caption.

List of Tables

List page numbers of all tables. The list should include a **short title** for each table but not the description of whole caption.

Chapter 1: Introduction

It should contain the following:

1.1 Background and statement of the problem (*should normally not exceed two pages for a B.Eng or a Master's dissertation*)

(Here, the project is introduced. Begin your introduction with broad statements that seek to situate the project idea within the broader area of science and technology. Then narrow down to a narration of what is known and what is not known (gaps in knowledge). The introduction section ends with a statement of the problem and what you plan to do to solve it (in the case of a project proposal) or have done to solve it (in the case of a project report).

A good introduction cannot be completed until the body of the dissertation (or proposal) is known. Consider completing the introductory section(s) after you have completed the rest of the project/dissertation (or proposal), rather than before.

1.2 Rationale (why you are carrying out the work)

1.3 Research questions or Research hypotheses (chose one only)

1.3.1 Main research question (corresponding to the overall objective)

1.3.2 Specific research questions or research hypotheses (corresponding to the specific objectives in nature and number) (chose one)

1.4 Objectives

1.4.1 Overall objective (singular tense)

1.4.2 Specific objectives (more than 1 but generally a maximum of four and numbered 1, 2, …4)

The overall objective must tie with the title of the project. The specific objectives are sub-actions you will undertake to address the major action mentioned in the overall of general objective. Action verbs should be used. For a proposal, use the future tense; For a report, use the past tense.

E.g.

In a Proposal: The overall objective of the present project is to……

In a Report: The overall objective of the present project was to……

In a Proposal: The specific objectives are to….

In a Project: The specific objectives were……

Literature review

Here, all the books, articles, dissertations, and every other written and published material related to the research topic are presented bringing out the findings, contributions and limitations of each. These findings are used to situate and bring out the relevance of the present work in the project/dissertation. This section seeks to review current knowledge in the area of study and identify gaps in knowledge, leading finally to the proposal on how the gaps could be filled.

Materials and Methods section

(subdivided into sections corresponding to the specific objectives)

What belongs in the "research methods" section of a project/dissertation?

1. Description of your theory, materials, procedure, analytical methods, including reference to any specialized statistical software.

2. Calculations, technique and calibration plots.

3. Limitations, assumptions, and range of validity.
4. Information needed by another researcher to replicate or reproduce your experiment/results.
5. Information to allow the reader to assess the validity of your research results.

Citations in this section should be limited to data sources and references of where to find more complete descriptions of procedures. Do not include descriptions of results here.

Results

(subdivided into sections corresponding to the specific objectives)

- The results are actual statements of observations, including statistics, tables and graphs.
- Indicate information on range of variation.
- Mention negative results as well as positive. Do not interpret results - save that for the discussion.
- Lay out the case as for a jury. Present sufficient details so that others can draw their own inferences and construct their own explanations.
- Use S.I. units (m, s, kg, W, etc.) throughout the thesis.
- Break up your results into logical segments by using subheadings
- Key results should be stated in clear sentences at the beginning of paragraphs. It is far better to say "X had significant positive relationship with Y (linear regression $p<0.01$, $r^2=0.79$)" then to start with a less informative like "There is a significant relationship between X and Y". Describe the nature of the findings; do not just tell the reader whether or not they are significant.

Quarantine your observations from your interpretations. The writer must make it crystal clear to the reader which statements are observation and which are interpretations. In most circumstances, this is best accomplished by physically separating statements about new observations from statements about the meaning or significance of those observations. Alternatively, this goal can be accomplished by careful use of phrases such as "I infer ...". For example, vast bodies of geological literature became obsolete with the advent of plate tectonics; the papers that survived are those in which observations were presented in stand-alone fashion, unmuddied by whatever ideas the author might have had about the processes that caused the observed phenomena.

How do you do this?

1. Physical separation into different sections or paragraphs.
2. Don't overlay interpretation on top of data in figures.
3. Careful use of phrases such as "We infer that ".

4. Don't worry if "results" seem short.

Why?

1. Easier for your reader to absorb, frequent shifts of mental mode not required.

2. Ensures that your work will endure in spite of shifting paradigms.

Discussion

Under this section, the student should essentially recall the problem statement, objectives, then in different paragraphs should recall a key result, interpret it, seek to justify it and relate it with previous findings in the literature. The discussion section should be a brief essay in itself, answering the following questions and caveats:

1. What are the major patterns in the observations? (Refer to spatial and temporal variations.)

2. What are the relationships, trends and generalizations among the results?

3. What are the exceptions to these patterns or generalizations?

4. What are the likely causes (mechanisms) underlying these patterns resulting predictions?

5. Is there agreement or disagreement with previous work?

6. Interpret results in terms of background laid out in the introduction - what is the relationship of the present results to the original question?

7. What is the implication of the present results for other unanswered questions in earth sciences, ecology, environmental policy, etc....?

8. Multiple hypotheses: There are usually several possible explanations for results. Be careful to consider all of these rather than simply pushing your favorite one. If you can eliminate all but one, that is great, but often that is not possible with the data in hand. In that case you should give even treatment to the remaining possibilities, and try to indicate ways in which future work may lead to their discrimination.

9. Avoid bandwagons: A special case of the above. Avoid jumping a currently fashionable point of view unless your results really do strongly support them.

10. What are the things we now know or understand that we didn't know or understand before the present work?

11. Include the evidence or line of reasoning supporting each interpretation.

12. What is the significance of the present results: why should we care?

This section should be rich in references to similar work and background needed to interpret results. Is there material that does not contribute to one of the elements listed above? Break up the section into logical segments by using subheads. Include elements of limitations of your study, if any.

Conclusions

(subdivided into sections corresponding to the specific objectives; followed by a general conclusion section)

- What is the strongest and most important statement that you can make from your observations?
- If you met the reader at a meeting six months from now, what do you want them to remember about your project/dissertation?
- Refer back to problem posed, and describe the conclusions that you reached from carrying out this investigation, summarize new observations, new interpretations, and new insights that have resulted from the present work.
- Include the broader implications of your results.

Recommendations

List the recommendations for improvement of the research you have done. Reflect on your results in isolation and in relation to what others have achieved in the same field. This self-analysis is particularly important.

References

The APA (American Psychological Association) Style shall be the referencing style used in all projects and dissertations in the National Higher Polytechnic Institute.

- cite all ideas, concepts, text, data that are not your own
- if you write a major statement, back it up with your own data or a reference
- all references cited in the main text must be listed in the References section. Also, only references cited in the main can be put in the References section.
- The APA style uses three kinds of information to be included in in-text citations. The **author's last name** and **date (year) of publication** must always appear, and these items must match exactly the corresponding entry in the references list. Eg …… (Cho-Ngwa, 2003); ……... Cho-Ngwa *et al.* (2018); …......(Cho-Ngwa and Nfah, 1995).
- cite single-author references by the surname of the author (followed by date of the publication in parenthesis)
    - ... according to Mofor (2017)…….
    - ... population growth is one of the greatest environmental concerns facing future generations (Hays, 1994).
- cite double-author references by the surnames of both authors (followed by date of the publication in parenthesis)
    - e.g. Fombu and Mofor (2020)…….
- cite more than double-author references by the surname of the first author followed by

et al. and then the date of the publication

- e.g. a paper published by Fidelis Cho-Ngwa, Eustace Nfah and Kennedy Fozao would be: Cho-Ngwa *et al.* (2020)……..
- If it is cited at the end of a statement, e.g. Other workers approached the problem through production and use of specific monoclonal antibodies (Bartlett *et al.,* 1975; Cabrera *et al.,* 1989).

- do not use footnotes
- Publications by the same author(s) in the same year shall be marked such as 2019a, 2019b, 2020a, 2020b, etc.
- In the reference list, journal articles are cited in the following order: Author(s) names, (Year of publication), name of journal in *Italics*, volume (number): page range. For example; Mbuh, M. K., Ngapgue, F., Kagou, D.A, Mofor, N. A., and Penka, J.B. (2020b). Modelling the thermomechanical behaviour of earth bricks stabilized with Portland cement. *International Journal of Advanced Research in Science, Engineering and Technology,* 7(10):15290-15301.
- In the reference list, published books are cited in the following order: Author(s) names, (Year of publication), titles of book in *Italics*, Edition, ISBN, Publisher, Country or Town of publication, Number of pages. For example; Benton, J.J.J. (2003). *Agronomic Handbook- Management of Crops, Soils, and Their Fertility*. 3rd Edtion, ISBN:0-8493-0897-6, CRC Press LLC., USA, Pp. 1035.
- For information obtained from the internet, follow the same procedure for books and add the search engine, date and time you accessed the document. For example; Waite, W.M. (1997). *Assessment of water supply and associated matters in relation to the incidence of cryptosporidiosis*, Drinking water inspectorate, http://www.doctoc.com/docs/8659019/thesis.htlm. Retrieved 15 May, 2014.
- The candidate shall ensure that the spelling of names and years are the same in the document and the list of references
- List all references cited in the text in **alphabetical order** using the following format for different types of material:

Benton, J.J.J. (2003). *Agronomic Handbook- Management of Crops, Soils, and Their Fertility*. ISBN:0-8493-0897-6, CRC Press LLC., USA, Pp. 1035.

Mbuh, M. K., Kagou, D., Ngapgue, F., Mofor, N. A., Manefouet, B. E. and Yamb , E. (2020a). Compressed Stabilized Earth Brick (CSEB) As Building Construction Elements. *IOSR Journal of Mechanical and Civil Engineering,* 17 (4):42-48.

Mbuh, M. K., Ngapgue, F., Kagou, D.A, Mofor, N. A., and Penka, J.B. (2020b). Modelling the thermomechanical behaviour of earth bricks stabilized with Portland cement. *International Journal of Advanced Research in Science, Engineering and Technology,* 7(10):15290-15301.

Mofor, N. A., Njoyim, E.B.T., Mbene, K., Yuhinwenkeh, N. B., and Nchofua, F. B. (2020).

Trace Element Status and Environmental Implications of Soils and *Zea mays* from Farmed Dumpsites in the Bamenda Metropolis, North-West Cameroon. *Hindawi Applied and Environmental Soil Science*, 2020:1-9.

Ngane, E.B., Tening, A.S., Ehabe, E.E. and Tchuenteu (2012). Potentials of some cement by products for liming of an acid soil in the humid zone of South-Western Cameroon. *Agriculture and Biology Journal of North America*, 3(8): 326-331.

Njoyim, E.B.T., Mvondo-Zé, A.D., Mofor, N.A. and Onana, A.A. (2016). Phosphorus adsorption isotherms in relation to soil characteristics of some selected volcanic affected soils of Foumbot in the West Region of Cameroon. *International Journal of Soil Science*, 11: 19-28.

Tellen, V.A. and Yerima, B.P.K. (2018). Efects of land use change on soil physicochemical properties in selected areas in the North West region of Cameroon. *Environmental Systems Research,* 7:1-29.

Waite, W.M. (1997). *Assessment of water supply and associated matters in relation to the incidence of cryptosporidiosis*, Drinking water inspectorate, http://www.doctoc.com/docs/8659019/thesis.htlm. Retrieved 15 May, 2014 at 5:00 pm.

SOME IMPORTANT ISSUES

1 **Plagiarism and Falsification of Data**: NAHPI expects students to follow the code of academic integrity of the University of Bamenda. They have the obligation to exhibit honesty and to respect the ethical standards of the academy as they carry out their academic assignments and must avoid plagiarism. Projects and Dissertations shall be checked for Plagiarism and violating students will be disciplined accordingly. Falsification of data shall be treated as a case of examination fraud.

2 **Ethical issues and presentation style of Projects and Dissertations:** A supervisor shall not write for a candidate nor shall the supervisor override the candidate's style by over-editing and prompting. The candidate shall ensure that the results are carefully recorded and calculations verified.

3 **Projects and Dissertations are being classified as publications.** Candidates hold copyright to their Dissertations and Theses regardless of whether they include a copyright page in them or register with the copyright office. Candidates will sign the UBa Library Projects and Dissertations consent form to protect their rights as authors of the research reports. However, candidates who do not wish that their works should be available for reasons such as pending patents, copyright agreements or future publication should seek advice from the Postgraduate School as to the restricted use or embargo.

4 **Other Provisions:** All postgraduate students shall consult the "Thesis and Dissertation Guide of The University of Bamenda" (UBa, 2018) for guidelines not defined in this document and keep track of modifications and amendments of the University of Bamenda Senate, through the administration of the National Higher Polytechnic Institute.

NB 1: *Samples of preliminary pages (cover page, title page, copyright page, declaration page, certification page and others) are presented on pages 14 to 21 below. These samples* ***must be respected*** *by all students.*

NB2: *The results of all NAHPI research projects/dissertations are owned by the University of Bamenda.*

FORMAT FOR RESEARCH PROPOSAL

i) Cover page

ii) Table of Contents

iii) Chapter 1: Introduction. It should contain the following:

1.5 Brief background of the study, statement of the problem

1.6 Rationale

1.7 Research questions or Research hypotheses (chose one).

1.7.1 Main research question (corresponding to the overall objective)

1.7.2 Specific research questions or research hypotheses (corresponding to the specific objectives in nature and number) (chose one)

1.8 Objectives

1.8.1 Overall objective (singular tense)

1.8.2 Specific objectives (more than 1 but generally a maximum of four and numbered 1, 2, …4)

1.9 Significance of the study

iv) Chapter 2: Literature review

v) Chapter 3: Methodology (subdivided into sections corresponding to the specific objectives)

vi) Chapter 4: Expected outcomes

vii) Time frame or schedule (subdivided into sections corresponding to the specific objectives; **Use a Gantt chart**)

viii) Budget estimate

ix) References

NB1: *The cover page of the research proposal should be the same as the cover page of the final project, but with* ***'Research proposal'*** *written just above the title of the project.*

NB2: *The total number of length for the research proposal in NAHPI should be maximum* ***10 pages.***

Cover page template

THE UNIVERSITY OF BAMENDA

NATIONAL HIGHER POLYTECHNIC INSTITUTE

DEPARTMENT OF ORIGIN

PROVIDE THE TITLE OF THE PROJECT

A Project Submitted to the Department of xxxxxx xxxxxx xxxxx in the National Higher Polytechnic Institute of The University of Bamenda in Partial Fulfillment of the Requirements for the Award of a Bachelor of Engineering Degree in xxxxxxx xxxxxxx

BY:

XXXXXXX XXXXXXXX XXXXXXXXX

REGISTRATION NUMBER: XXXXXXXXXXXXX

SUPERVISOR(S):

XXXXXX XXXXXXXX *(NAME OF SUPERVISOR)*
(*Academic rank of supervisor*)

XXXXXX XXXXXXXX *(NAME OF CO-SUPERVISOR, if any)*
(*Academic rank of co-supervisor*)

MONTH, YEAR

Title page template

THE UNIVERSITY OF BAMENDA

NATIONAL HIGHER POLYTECHNIC INSTITUTE

DEPARTMENT OF ORIGIN

PROVIDE THE TITLE OF THE PROJECT

A Project Submitted to the Department of xxxxxx xxxxxx xxxxx in the National Higher Polytechic Institute of The University of Bamenda in Partial Fulfillment of the Requirements for the Award of a Bachelor of Engineering Degree in xxxxxxx xxxxxxx

BY:

XXXXXXX XXXXXXXX XXXXXXXXX

REGISTRATION NUMBER: XXXXXXXXXXXXX

SUPERVISOR(S):

XXXXXX XXXXXXXX *(NAME OF SUPERVISOR)*
(*Academic rank of supervisor*)

XXXXXX XXXXXXXX *(NAME OF CO-SUPERVISOR, if any)*
(*Academic rank of co-supervisor*)

MONTH, YEAR

DECLARATION OF ORIGINALITY OF STUDY

I, xxxxxxxx xxxxxx xxxxxxxx, registration N◦ : xxxxxxxxxxx, in the Department of xxxxxxxx xxxxxxxxxx xxxxxxxxxx, National Higher Polytechnic Institute, The University of Bamenda hereby declare that this work titled “xxxxxxxxxxx xxxxxxxxxx xxxxxx xxxxxxxxxxxxx” is my original work. It has not been presented in any application for a degree or any academic pursuit elsewhere. I have acknowledged all borrowed ideas nationally and internationally through citations.

Date: ___________________ Signature of author ________________

CERTIFICATION OF CORRECTIONS AFTER DEFENCE

This is to certify that this project titled “xxxxxxxxxx xxxxxxxx xxxxxxxx xxxx xxxxxxx” is the original work of xxxxxxx xxxxxxxx xxxxxx. This work is submitted in partial fulfillment of the requirements for the award of a Bachelor of Engineering Degree in xxxxxxx xxxxxxx in the National Higher Polytechnic Institute of The University of Bamenda, Cameroon.

Supervisor : ____
Engr./Dr./Prof. Xxxxxxxx xxxxxxxx

The Head of Department : ____
Engr./Dr./Prof. Xxxxx xxxxxx xxxxxxx

The Director : ____
Prof. Xxxxx xxxxxx

ACCEPTANCE OF DISSERTATION (applies to Master programs only)

Having met the stipulated requirements, this dissertation entitled “xxxxxxxxxxx xxxxxxxxx xxxxxxx xxxxxxx” has been accepted by the Postgraduate School of The University of Bamenda in partial fulfillment of the requirements for the award of a Master of Engineering Degree in xxxxxxx xxxxxxx (Option: xxxxxxxxx xxxxxxx (if any)).

Chairperson, Master’s Dissertation Committee : __________________________
Prof. Xxxxxxxx xxxxxxxx

Date : ____________________________

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Attestation of Correction of Thesis/Dissertation/Project

This is to attest that:

Name:____

Registration Number:____

Department:____

Option:____

Faculty/School:____

Title:____

____

____

Has effected corrections identified during the defense of his/her work as recommended.

Examiner(s) Supervisor(s) Head of Department

________________ ________________ ________________

Date……………. Date………………. Date……………….

THESES/DISSERTATIONS SUBMISSION FORM

The following student,

Name:…………………………………………………………………….........

Registration N^{o}:……………………………………………………………….

Faculty/School:………………………………………………………………..

Department:……………………………………………………………………

is authorized to submit his/her thesis/dissertation at the Central Library for binding.

Head of Department

ADDITIONAL NOTES

2.3 Details of the elements

2.3.1 Project Title

Project Title is **a name of the Project**. A proper project title describes the whole assignment in one sentence. Therefore:

- it must be succinctly written,
- catchy and generate interest to a passers-by,
- contains no jargon or unfamiliar abbreviations,
- should be 5-15 words (usually never more than 20 words);
- excellent grammar

2.3.2 Abstract

- A good abstract
  - begins with a statement of the problem that explains in 1-3 sentences the problem the project is being conducted to resolve. For a research proposal, this segment could be much longer.
  - It then goes on to state the purpose, goal or overall objective of the project.
  - It moves on with elements of the project design, methodology, results obtained (or expected results in the case of a proposal); and the major conclusion (in the case of a dissertation or work already done).
  - is concise, succinct, readable, excellent grammar, quantitative and contains all the keywords.

- Answers to these questions should be found in the abstract:
  1. Why did you do it? What question were you trying to answer?
  2. How did you do it? State methods.
  3. What did you learn? State major results.
  4. Why does it matter? Make a conclusion statement that captures the major results.
- Abstracts do not have citations.

An examples of a good abstracts with the different colours showcasing the various components:

Title: Genetic Relationship Between the Different Ethno-Linguistic Groups of Mezam and Momo Divisions of the North West Region of Cameroon

Abstract:

Onchocerciasis is a neglected tropical disease that currently afflicts an estimated 15 million people and is the second leading infectious cause of blindness world-wide. The development of a macrofilaricide to cure the disease has been hindered by the lack of appropriate small laboratory animal models. This study therefore, was aimed at developing and validating the Mongolian gerbil, as an *Onchocerca* adult male worm model. The gerbils were each implanted with 20 *O. ochengi* male worms (collected from infected cattle), in the peritoneum. Following drug or placebo treatments, the implanted worms were recovered from the animals and analyzed for burden, motility and viability. Worm recovery in control gerbils was on average 35%, with 89% of the worms being 100% motile. Treatment of the gerbils implanted with male worms with flubendazole (FBZ) resulted in a significant reduction ($p = 0.0021$) in worm burden (6.0% versus 27.8% in the control animals); all recovered worms from the treated group had 0% worm motility versus 91.1% motility in control animals. FBZ treatment had similar results even after four different experiments. Using this model, we tested a related drug, oxfendazole (OFZ), and found it to also significantly ($p = 0.0097$) affect worm motility (22.7% versus 95.0% in the control group). We conclude that we have successfully developed and validated a novel gerbil *O. ochengi* adult male worm model for testing new macrofilaricidal drugs in *vivo.*

Keywords: Onchocerciasis, animal model, drug screens

Title:

Abstract:

Ultralight bosons such as axion-like particles are viable candidates for dark matter. They can form stable, macroscopic field configurations in the form of topological defects that could concentrate the dark matter density into many distinct, compact spatial regions that are small compared with the Galaxy but much larger than the Earth. Here we report the results of the search for transient signals from the domain walls of axion-like particles by using the global network of optical magnetometers for exotic (GNOME) physics searches. We search the data, consisting of correlated measurements from optical atomic magnetometers located in laboratories all over the world, for patterns of signals propagating through the network consistent with domain walls. The analysis of these data from a continuous month-long operation of GNOME finds no statistically significant signals, thus placing experimental constraints on such dark matter scenarios.

Keywords:

2.3.3 Introduction

This section introduces the readers to the problem the project is trying to solve or the questions that are to be answered. In arranges the statements in a funnel-shape structure, from general and broader statements to more and more focused statements, ending up in the purpose of the project in the early paragraph.

A good introduction can't be completed until the body of the thesis (or of the project proposal) is known. Consider completing the introductory section(s) after you have completed the rest of the project/dissertation (or project proposal), rather than before. The Introduction finally ends with a statement recapitulating the main research question, followed by the solution being set forward (in the case of proposal) or that has been set forward (in the case of a thesis) to address it.

An example of a good introduction follows:

MARIDA: A benchmark for Marine Debris detection from Sentinel-2 remote sensing data

Introduction

Marine Debris, such as plastics, is a major global issue with important environmental, economic, human health and aesthetic aspects. Plastics remain in the ocean for a long time, and have been found in various areas worldwide [1–3], affecting marine life at different trophic levels [4]. To tackle the Marine Debris issue, several solutions for detecting [5, 6], cleaning [7] and preventing [8] have been developed and validated. Among those, detecting and monitoring floating litter has recently gained the attention of most research and development efforts [9].

In particular, earth observation data from public and commercial satellite programs [10–14] have been employed for detecting and monitoring Marine Debris, as well as remote sensing data from manned aircraft [15], unmanned aerial vehicles (UAVs) [16–20], bridge-mounted [21] and underwater-cameras [22]. Spectral indices have also been proposed to enhance the detection of Marine Debris on multispectral satellite data, like the Floating Debris Index (FDI) [13] and the Plastic Index (PI) [23] that have been developed based on artificial plastic targets.

Furthermore, to better understand the spectral behaviour of Marine Debris, hyperspectral measurements have been conducted, exploring sensors' capabilities in distinguishing plastics from other features such as vegetation, natural material, and water types [24–28]. Investigating Marine Debris characteristics (including its spectral behavior) has been also attempted via multispectral satellite observations [10, 12, 13, 29], highlighting that spectral discrimination of Marine Debris from other sea surface features (e.g., ships, foam) is not straightforward. Indeed, differentiating floating plastic debris from bright features, such as waves, sunglint, clouds, is currently considered very challenging [5, 6]. This is due to the fact that plastics have complex properties, diversifying in color, chemical composition, size and level of water submersion [30, 31]. A high-quality dataset can address the challenges mentioned above, supporting also the development and improvement of Marine Debris detection methods, and assessing the operational aspects of any given solution (e.g., scalability).

However, despite the challenging and continuously growing issue of Marine Debris, the currently available datasets are relatively limited in number and do not usually employ open-access high-resolution satellite data over geographically extended areas. These facts prohibit satellite data exploitation from ML frameworks and operational solutions. In addition, most of the currently available marine remote sensing datasets focus on detecting specific objects such as vessels [32–35]. Datasets for cloud detection over the ocean [36] and *Sargassum* macroalgae extraction [37, 38] have also been developed with a limited number of classes.

To this end, this study aims to fill this gap with a new, open-access benchmark dataset, named MARIDA—MARIne Debris Archive, based on S2 multispectral satellite data. MARIDA offers real cases with Marine Debris events, providing globally distributed annotations, ready for ML tasks. The produced dataset takes an innovative step forward by containing sea features that co-exist in remote sensing images, ultimately forming 15 thematic classes in total. Along with MARIDA, ML baselines for the weakly supervised semantic segmentation task [39] are presented, including shallow ML and deep neural network architectures. To enlarge the benchmark application area, the multi-label classification task is also considered.

2.3.4 Literature review

A literature review is a survey of scholarly sources on a specific topic. It provides an overview of current knowledge, allowing you to identify relevant theories, methods, and gaps in the existing research.

Writing a literature review involves finding relevant publications (such as books and journal articles), critically analyzing them, and explaining what you found. There are five key steps:

1. **Search** for relevant literature
2. **Evaluate** sources
3. **Identify** themes, debates and gaps
4. **Outline** the structure
5. **Write** your literature review

A good literature review doesn't just summarize sources—it analyzes, synthesizes, and critically evaluates to give a clear picture of the state of knowledge on the subject.

In writing the literature review:

- Be chronological
- Start by reviewing the problem and scope and defining the key terms
- Ensure all important concepts or topics around the project have been reviewed in separate headings or subheadings
- Avoid plagiarism or any extensive copying without giving credits to the source
- Avoid citing publications you have not read

This section seeks to review current knowledge in the area of study and identify gaps in knowledge, leading finally to the proposal on how the gaps could be filled.

2.3.5 Problem Statement / Project rationale

A research problem is an area of concern or a gap in the existing knowledge that points to the need for further understanding and investigation. A problem statement is used in research work as a claim that outlines the problem addressed by a study. The problem statement briefly explains the problem that the research will address. If the topic of your research is food safety in the school feeding system, you need to first identify why food safety is lacking in schools. Your problem statement can explain that food safety in school feeding systems is an important concern and point towards a gap in research that shows that this problem has not been addressed. Adapted from: https://www.editage.com/insights/how-to-write-a-problem-statement-for-my-research Retrieved January 08, 2022

The rationale of your research is the reason for conducting the study. The rationale should answer the need for conducting the said research. It is a very important part of your write-up as it justifies the significance and novelty of the study. That is why it is also referred to as the justification of the study.

To write your rationale, you should first write a background on what all research has been done on your study topic. Follow this with 'what is missing' or 'what are the open questions of the study'. Identify the gaps in the literature and emphasize why it is important to address those gaps. This will form the rationale of your study. The rationale should be followed by a hypothesis and objectives.

The rationale of a thesis is usually written on about half to a full A4 page.

From the forgoing, it is clear that both the Problem Statement (or Statement of the Problem) and the Rationale are addressing the same question of: Why does this research need to be conducted?

For conference presentations, the statement of the problem is generally incorporated into the introduction; academic proposals for theses or dissertations should have this as a separate section.

Adapted from https://www.fundsforngos.org/how-to-write-a-proposal/writing-problem-statementproject-rationale-in-a-proposal/ Retrieved January 08, 2022

2.3.6 Research questions or research hypotheses

Research Questions are relevant to normative or census type research (How many of them are there? Is there a relationship between them?). They are most often used in qualitative inquiry. *Hypotheses* are relevant to theoretical research and are typically used only in quantitative inquiry.

A *research question* poses a relationship between two or more variables but phrases the relationship as a question; a *hypothesis* represents a declarative statement of the relations between two or more variables

Deciding whether to use questions or hypotheses depends on factors such as the purpose of the study, the nature of the design and methodology, and the audience of the research (at times even the taste and preference of committee members, particularly the Chair).

Hypotheses can be null or alternate

Example of *null*— “There is no difference in school achievement for high and low self-regulated students.”

Example of *alternative*— “High self-regulated students will achieve more in their classes than low self-regulated students.”

In general, the null hypothesis is used if theory/literature does not suggest a hypothesized relationship between the variables under investigation; the alternative is generally reserved for situations in which theory/research suggests a relationship or directional interplay.

2.3.7 Overall and specific objectives

2.3.7.1 Overall objective (also called aim or overall goal):

- **Usually starts like this:** The overall objective of the present study is to (or was to)…… (in the case where the work has already been done, use the past tense)
- State clearly and succinctly the overall objective of the study
- This must reflect and recapitulate the keywords of the project title – use the project title to formulate the overall objective.
- This must be in singular tense

2.3.7.2 Specific objectives (also called objectives):

- **Usually starts like this:** The specific objectives were:……
  1…..
  2…..
  etc
- Number the specific objectives (1, 2, 3…)
- They should be at least two, but should generally not be more than 4, unless exceptionally
- The rest of the chapters must then be arranged into sections following the specific objectives. Thus, if there are 3 specific objectives, the Materials and Methods or Methodology section would be arranged in 3 main sections, and the results also in 3 main sections corresponding to the 3 specific objectives.

2.3.7.3 Example of overall and specific objectives:

Objectives

Overall objective

The overall objective of this study was to determine the genetic relationship between the different ethno-linguistic groups of Mezam and Momo Divisions of the North West Region of Cameroon

Specific objectives

The specific objectives were:

1. To determine the allele frequency distribution of each allele in this study;
2. To determine the number of haplotypes in these populations;
3. To establish the genetic relationship between the different linguistic groups;
4. To establish any correlation between linguistic relatedness and genetic relatedness among these populations

2.3.8 Significance of study

Indicate how your research will refine, revise, or extend existing knowledge in the area under investigation. Note that such refinements, revisions, or extensions may have either substantive, theoretical, or methodological significance. Think pragmatically (i.e., cash value).

Most studies have two potential audiences: practitioners and professional peers. Statements relating the research to both groups are in order.

This can be a difficult section to write. Think about *implications*—how results of the study may affect scholarly research, theory, practice, educational interventions, curricula, counseling, policy.

When thinking about the significance of your study, ask yourself the following questions.

1. What will results mean to the theoretical framework that framed the study?
2. What suggestions for subsequent research arise from the findings?
3. What will the results mean to the practicing educator?
4. Will results influence programs, methods, and/or interventions?
5. Will results contribute to the solution of educational problems?
6. Will results influence educational policy decisions?
7. What will be improved or changed as a result of the proposed research?
8. How will results of the study be implemented, and what innovations will come about?

2.3.9 Limitations and Delimitations of study

A *limitation* identifies potential weaknesses of the study. Think about your analysis, the nature of self-report, your instruments, the sample. Think about threats to internal validity that may have been impossible to avoid or minimize—explain.

A *delimitation* addresses how a study will be narrowed in scope, that is, how it is bounded. This is the place to explain the things that you are not doing and why you have chosen not to do them—the literature you will not review (and why not), the population you are not studying (and why not), the methodological procedures you will not use (and why you will not use them). Limit your delimitations to the things that a reader might reasonably expect you to do but that you, for clearly explained reasons, have decided not to do.

2.3.10 Project Design and Methodology, Materials and Methods

(see Chapter 1 for details)

Methodology is usually used in the research proposal, while Materials and Methods, which are the details of the Methodology, are mostly used in reporting the research.

2.3.11 Expected outcomes (*in proposals only*)

In your proposal write-up, give a structured narration of the expected outcomes of your study in the place of results, since you don’t have results yet.

2.3.12 Results

The results section of the research paper is where you report the findings of your study based upon the information gathered as a result of the methodology [or methodologies] you applied. The results section should simply state the findings, without bias or interpretation, and arranged in a logical sequence. The results section should always be written in the past tense. A section describing results [a.k.a., "findings"] is particularly necessary if your paper includes data generated from your own research.

Importance of a Good Results Section

When formulating the results section, it's important to remember that the results of a study do not prove anything. Research results can only confirm or reject the research problem underpinning your study. However, the act of articulating the results helps you to understand the problem from within, to break it into pieces, and to view the research problem from various perspectives.

The page length of this section is set by the amount and types of data to be reported. Be concise, using non-textual elements, such as figures and tables, if appropriate, to present results more effectively. In deciding what data to describe in your results section, you must clearly distinguish material that would normally be included in a research paper from any raw data or other material that could be included as an appendix. In general, raw data should not be included in the main text of your paper unless requested to do so by your professor.

Avoid providing data that is not critical to answering the research question. The background information you described in the introduction section should provide the reader with any additional context or explanation needed to understand the results. A good rule is to always re-read the background section of your paper after you have written up your results to ensure that the reader has enough context to understand the results [and, later, how you interpreted the results in the discussion section of your paper].

Structure and Writing Style

I. Structure and Approach

For most research paper formats, there are two ways of presenting and organizing the results.

1. **Present the results followed by a short explanation of the findings**. For example, you may have noticed an unusual correlation between two variables during the analysis of your findings. It is correct to point this out in the results section. However, speculating as to why this correlation exists, and offering a hypothesis about what may be happening, belongs in the discussion section of your paper.
2. **Present a section and then discuss it, before presenting the next section then discussing it, and so on**. This is more common in longer papers because it helps the reader to better understand each finding. In this model, it can be helpful to provide a brief conclusion in the results section that ties each of the findings together and links to the discussion.

NOTE: The discussion section should generally follow the same format chosen in presenting and organizing the results.

II. Content

In general, the content of your results section should include the following elements:

1. An introductory context for understanding the results by restating the research problem that underpins the purpose of your study.
2. A summary of your key findings arranged in a logical sequence that generally follows your methodology section.
3. Inclusion of non-textual elements, such as, figures, charts, photos, maps, tables, etc. to further illustrate the findings, if appropriate.
4. In the text, a systematic description of your results, highlighting for the reader observations that are most relevant to the topic under investigation [remember that not all results that emerge from the methodology that you used to gather the data may be relevant].
5. Use of the past tense when refering to your results.

6. The page length of your results section is guided by the amount and types of data to be reported. However, focus only on findings that are important and related to addressing the research problem.

Using Non-textual Elements

- Either place figures, tables, charts, etc. within the text of the result, or include them in the back of the report--do one or the other but never do both.
- In the text, refer to each non-textual element in numbered order [e.g., Table 1, Table 2; Chart 1, Chart 2; Map 1, Map 2].
- If you place non-textual elements at the end of the report, make sure they are clearly distinguished from any attached appendix materials, such as raw data.
- Regardless of placement, each non-textual element must be numbered consecutively and complete with caption [caption goes under the figure, table, chart, etc.]
- Each non-textual element must be titled, numbered consecutively, and complete with a heading [title with description goes above the figure, table, chart, etc.].
- In proofreading your results section, be sure that each non-textual element is sufficiently complete so that it could stand on its own, separate from the text.

III. Problems to Avoid

When writing the results section, avoid doing the following:

1. **Discussing or interpreting your results**. Save all this for the next section of your paper, although where appropriate, you should compare or contrast specific results to those found in other studies [e.g., "Similar to Smith [1990], one of the findings of this study is the strong correlation between motivation and academic achievement...."].
2. **Reporting background information or attempting to explain your findings**; this should have been done in your Introduction section, but don't panic! Often the results of a study point to the need to provide additional background information or to explain the topic further, so don't think you did something wrong. Revise your introduction as needed.
3. **Ignoring negative results**. If some of your results fail to support your hypothesis, do not ignore them. Document them, then state in your discussion section why you believe a negative result emerged from your study. Note that negative results, and how you handle them, often provides you with the opportunity to write a more engaging discussion section, therefore, don't be afraid to highlight them.
4. **Including raw data or intermediate calculations**. Ask your professor if you need to include any raw data generated by your study, such as transcripts from interviews or data files. If raw data is to be included, place it in an appendix or set of appendices that are referred to in the text.
5. **Be as factual and concise as possible in reporting your findings**. Do not use phrases that are vague or non-specific, such as, "appeared to be greater or lesser than..." or "demonstrates promising trends that...."
6. **Presenting the same data or repeating the same information more than once**. If you feel the need to highlight something, you will have a chance to do that in the discussion section.
7. **Confusing figures with tables**. Be sure to properly label any non-textual elements in your paper. If you are not sure, look up the term in a dictionary.

Data analysis

Specify the procedures you will use, and label them accurately (e.g., ANOVA, MANCOVA, HLM, ethnography, case study, grounded theory). Communicate your precise intentions and reasons for choosing specific tools. This helps you and the reader evaluate the choices you made and procedures you followed.

Provide a well thought-out rationale for your decision to use the design, methodology, and analyses you have selected.

Data presentation and display

For Vector Graphics (high resolution graphics)

Software:

QtiPlot (free), a clone of OriginLab (used for example in Pharma)

GIMP (free) an alternative of Adobe Photoshop

FIJI ImageJ

InkScape (free), a clone of Adobe Illustrator, CorelDraw

2.3.13 Discussion

The discussion section should include the following:

I. Reiterate the Research Problem/State the Major Findings

Briefly reiterate for your readers the research problem or problems you are investigating and the methods you used to investigate them, then move quickly to describe the major findings of the study. You should write a direct, declarative, and succinct proclamation of the study results.

II. Explain the Meaning of the Findings and Why They are Important

No one has thought as long and hard about your study as you have. Systematically explain the meaning of the findings and why you believe they are important. After reading the discussion section, you want the reader to think about the results ["why hadn't I thought of that?"]. You don't want to force the reader to go through the paper multiple times to figure out what it all means. Begin this part of the section by repeating what you consider to be your most important finding first.

III. Relate the Findings to Similar Studies

No study is so novel or possesses such a restricted focus that it has absolutely no relation to other previously published research. The discussion section should relate your study findings to those of other studies, particularly if questions raised by previous studies served as the motivation for your study, the findings of other studies support your findings [which strengthens the importance of your study results], and/or they point out how your study differs from other similar studies.

IV. Consider Alternative Explanations of the Findings

It is important to remember that the purpose of research is to *discover* and not to *prove*. When writing the discussion section, you should carefully consider all possible explanations for the study results, rather than just those that fit your prior assumptions or biases.

V. Acknowledge the Study's Limitations

It is far better for you to identify and acknowledge your study's limitations than to have them pointed out by your professor! Describe the generalizability of your results to other situations, if applicable to the method chosen, then describe in detail problems you encountered in the method(s) you used to gather information. Note any unanswered questions or issues your study did not address, and....

VI. Make Suggestions for Further Research

Although your study may offer important insights about the research problem, other questions related to the problem likely remain unanswered. Moreover, some unanswered questions may have become more focused because of your study. You should make suggestions for further research in the discussion section.

NOTE: Besides the literature review section, the majority of references to sources in your research paper are usually found in the discussion section. A few historical references may be helpful for perspective but most of the references should be relatively recent and included to aid in the interpretation of your results and/or linked to similar studies. If a study that you cited disagrees with your findings, don't ignore it--clearly explain why the study's findings differ from yours.

Problems to Avoid

- **Do not waste entire sentences restating your results**
- **Do not introduce new results in the discussion**
- **Use of the first person is acceptable, but** too much use of the first person may actually distract the reader from the main points.

NB: You can arrange your Discussion section materials according to your specific objectives. But these should be followed by a section that covers a recapitulation of the reasons for the study, the key finding and its justification and introduces the rest of the section.

2.3.14 Conclusions

Group the points in your conclusions according to your specific objectives and number likewise.

The conclusion should begin from the main research question that your thesis or dissertation aimed to address. This is your final chance to show that you've done what you set out to do, so make sure to formulate a clear, concise answer.

Don't repeat a list of all the results that you already discussed, but synthesize them into final takeaway statements that the reader will remember.

Presentation of Research Findings in Defences and Scientific Events

Tips for Making Effective PowerPoint Presentations

Adapted from: https://www.ncsl.org/legislators-staff/legislative-staff/legislative-staff-coordinating-committee/tips-for-making-effective-powerpoint-presentations.aspx Retrieved January 8, 2022.

Slideshows are quick to produce, easy to update and an effective way to inject visual interest into almost any presentation.

However, slideshows can also spell disaster even for experienced presenters. The key to success is to make certain your slideshow is a visual aid and not a visual distraction.

Tips for Making Effective PowerPoint Presentations

- Use the **slide master** feature to create a consistent and simple design template. It is fine to vary the content presentation (bulleted list, two-column text, text and image, etc.), but be consistent with other elements such as font, colors and background.
- **Limit your slides to about five lines of text each**
- **Aim for a minimum of 24-40 point font size**
- Simplify and limit the number of words on each screen. Use key phrases and include only essential information.
- Limit punctuation and avoid putting words in all-capital letters. Empty space on the slide will enhance readability.
- Use contrasting colors for text and background. Light text on a dark background is best. Patterned backgrounds can reduce readability.
- Avoid the use of flashy transitions such as text fly-ins. These features may seem impressive at first but are distracting and get old quickly.
- Overuse of special effects such as animation and sounds may make your presentation "cutesy" and could negatively affect your credibility.
- Use good-quality images that reinforce and complement your message. Ensure that your image maintains its impact and resolution when projected on a larger screen.

- Limit the speed of changing slides. Presenters who constantly “flip” to the next slide are likely to lose their audience. A good rule of thumb is one slide per minute.

- If possible, view your slides on the screen you’ll be using for your presentation. Make sure the slides are readable from the back row seats. Text and graphic images should be large enough to read but not so large as to appear “loud.”

- Have a Plan B in the event of technical difficulties. Remember that transparencies and handouts will not show animation or other special effects.

- Practice with someone who has never seen your presentation. Ask them for honest feedback about colors, content and any effects or graphic images you’ve included.

- Do not read too much from your slides. The content of your slides is for the audience, not for the presenter.

- Do not speak to your slides. Many presenters face their presentation onscreen rather than their audience.

- Do not apologize for anything in your presentation. If you believe something will be hard to read or understand, don’t use it.

The Seven Deadly Sins of PowerPoint Presentations

The key to success is to make certain your slide show is a visual aid and not a visual distraction. For the best results, avoid these common “seven deadly sins” of PowerPoint presentations.

1. **Slide Transitions and Sound Effects:** Transitions and sound effects can become the focus of attention, which in turn distracts the audience. Worse yet, when a presentation containing several effects and transitions runs on a computer much slower than the one on which it was created, the result is a sluggish, almost comical when viewed. Such gimmicks rarely enhance the message you’re trying to communicate. Unless you are presenting at a science fiction convention, leave out the laser-guided text! Leave the fade-ins, fade-outs, wipes, blinds, dissolves, checkerboards, cuts, covers and splits to Hollywood filmmakers.

2. **Standard Clipart:** Death to screen beans! PowerPoint© is now so widely used the clipart included with it has become a “visual cliché.” It shows a lack of creativity and a tired adherence to a standard form. First, make certain that you need graphical images to enhance your message. If you do, use your own scanned photographs or better-quality graphics from companies such as PhotoDisc (www.photodisc.com) or Hemera’s Photo Objects (www.hemera.com). Screen captures can add realism when presenting information about a Website or computer program.

3. **Presentation Templates:** Another visual cliché. Templates force you to fit your original ideas into someone else’s pre-packaged mold. The templates often contain distracting backgrounds and poor color combinations. Select a good book on Web graphics and apply the same principles to your slides. Create your own distinctive look or use your company logo in a corner of the screen.

4. **Text-Heavy Slides:** Projected slides are a good medium for depicting an idea graphically or providing an overview. Slides are a poor medium for detail and reading. Avoid paragraphs, quotations and even complete sentences. **Limit your slides to about five lines of text each** and use words and phrases to make your points. The audience will be able to digest and retain key points more easily. Don't use your slides as speaker's notes or to simply project an outline of your presentation.

5. **The "Me" Paradigm:** Presenters often scan a table or graphical image directly from their existing print corporate material and include it in their slide show presentations. The results are almost always sub-optimal. Print visuals are usually meant to be seen from 8-12 inches rather than viewed from several feet. Typically, these images are too small, too detailed and too textual for an effective visual presentation. The same is true for font size; 12 point font is adequate when the text is in front of you. In a slideshow**, aim for a minimum of 24-40 point font size**. Remember the audience and move the circle from "me" to "we." Make certain all elements of any particular slide are large enough to be seen easily. Size really does matter.

6. **Reading:** A verbal presentation should focus on interactive speaking and listening, not reading by the speaker or the audience. The demands of spoken and written language differ significantly. Spoken language is shorter, less formal and more direct. Reading text ruins a presentation. A related point has to do with handouts for the audience. One of your goals as a presenter is to capture and hold the audience's attention. If you distribute materials before your presentation, your audience will be reading the handouts rather than listening to you. Often, parts of an effective presentation depend on creating suspense to engage the audience. If the audience can read everything you're going to say, that element is lost.

7. **Faith in Technology:** You never know when an equipment malfunction or incompatible interfaces will force you to give your presentation on another computer. Be prepared by having a back-up of your presentation on a CD-ROM. Better yet is a compact-flash memory card with an adapter for the PCMCIA slot in your notebook. With it, you can still make last-minute changes. It's also a good idea to prepare a few color transparencies of your key slides. In the worst-case scenario, none of the technology works and you have no visuals to present. You should still be able to give an excellent presentation if you focus on the message. Always familiarize yourself with the presentation, practice it and be ready to engage the audience regardless of the technology that is available. It's almost a lost art.

Tips for Effective PowerPoint Presentations

Fonts

- Select a single sans-serif fonts such as Arial or Helvetica. Avoid serif fonts such as Times New Roman or Palatino because these fonts are sometimes more difficult to read.
- Use no font size smaller than 24 point.
- Use the same font for all your headlines.

- Select a font for body copy and another for headlines.
- Use bold and different sizes of those fonts for captions and subheadings.
- Add a fourth font for page numbers or as a secondary body font for sidebars.
- Don't use more than four fonts in any one publication.
- Clearly label each screen. Use a larger font (35-45 points) or different color for the title.
- Use larger fonts to indicate importance.
- Use different colors, sizes and styles (e.g., bold) for impact.
- Avoid italicized fonts as these are difficult to read quickly.
- Avoid long sentences.
- Limit punctuation marks.
- No more than 6-8 words per line
- For bullet points, use the 6 x 6 Rule. One thought per line with no more than 6 words per line and no more than 6 lines per slide
- Use dark text on light background or light text on dark background. However, dark backgrounds sometimes make it difficult for some people to read the text.
- Do not use all caps except for titles.
- Put repeating elements (like page numbers) in the same location on each page of a multi-page document.
- To test the font, stand six feet from the monitor and see if you can read the slide.

Design and Graphical Images

- Use design templates.
- Standardize position, colors, and styles.
- Include only necessary information.
- Limit the information to essentials.
- Content should be self-evident
- Use colors that contrast and compliment.
- Too may slides can lose your audience.
- Keep the background consistent and subtle.
- Limit the number of transitions used. It is often better to use only one so the audience knows what to expect.
- Use a single style of dingbat for bullets throughout the page.
- Use the same graphical rule at the top of all pages in a multi-page document.
- Use one or two large images rather than several small images.
- Prioritize images instead of a barrage of images for competing attention.
- Make images all the same size.
- Use the same border.
- Arrange images vertically or horizontally.
- Use only enough text when using charts or graphical images to explain the chart or graph and clearly label the image.
- Keep the design clean and uncluttered. Leave empty space around the text and graphical images.
- Use quality clipart and use it sparingly. A graphical image should relate to and enhance the topic of the slide.

- Try to use the same style graphical image throughout the presentation (e.g., cartoon, photographs)
- Limit the number of graphical images on each slide.
- Repetition of an image reinforces the message. Tie the number of copies of an image to the numbers in your text.
- Resize, recolor, reverse to turn one image into many. Use duplicates of varying sizes, colors, and orientations to multiply the usefulness of a single clip art image.
- Make a single image stand out with dramatic contrast. Use color to make a dramatic change to a single copy of your clip art.
- Check all images on a projection screen before the actual presentation.
- Avoid flashy images and noisy animation effects unless it relates directly to the slide.

Color

- Limit the number of colors on a single screen.
- Bright colors make small objects and thin lines stand out. However, some vibrant colors are difficult to read when projected.
- Use no more than four colors on one chart.
- Check all colors on a projection screen before the actual presentation. Colors may project differently than what appears on the monitor.

General Presentation

- Plan carefully.
- Do your research.
- Know your audience.
- Time your presentation by practicing with it several times.
- Speak comfortably and clearly.
- **It is normal to fear, shiver, stammer and sweat. However, always remember to stay focused on passing on the message clearly.**
- Check the spelling and grammar.
- Do not read the presentation. Practice the presentation so you can speak from bullet points. The text should be a cue for the presenter rather than a message for the viewer.
- Give a brief overview at the start. Then present the information. Finally review important points.
- It is often more effective to have bulleted points appear one at a time so the audience listens to the presenter rather than reading the screen.
- Use a wireless mouse or pick up the wired mouse so you can move around as you speak.
- If the content is complex, print the slides so the audience can take notes.
- **Do not turn your back on the audience**. Try to position the monitor so you can speak from it.